Alexandre Silva
Bachelor's Student in Computer Science and Researcher
UNIVEM - Centro Universitário Eurípides de Marília
Rua Hygino Muzy Filho,529 - Mirante
Marília - SP, 17525-901
alexandresilvaunivem@gmail.com
18/12/2024

Editor-in-Chief
Acta Informatica
Springer

Dear Henning Fernau,

I am pleased to submit our manuscript entitled "Quantum Oracles: How to Transform Classical Problems Into Quantum" for consideration for publication in Acta Informatica. Our work explores the uses of quantum oracles, describing their applications and tackling some problems with them. We believe that this work is a significant contribution to the Quantum Computing Community, especially for new researchers as a source of knowledge and inspiration.

Our study provides the fundamental components of many quantum algorithms and key insights into how to encode data in quantum circuits. For instance, in this paper, we explored some adjustments to Grover’s algorithm, testing whether different superpositions could yield better results during search routines. Also, a quantum game was implemented using oracles as the player's strategy, applying a classical optimization process to refine his strategy, thus maximizing the player’s winnings. In total, five problems were solved, demonstrating how these quantum techniques can be useful—or not—for some specific cases.

This manuscript is an original work that has not been published elsewhere and is not under consideration by another journal. All authors have reviewed and approved the final manuscript and have no conflicts of interest to declare.

We believe that this work aligns closely with the scope of Acta Informatica, particularly in: Quantum Computing, Algorithms and their analysis, Data handling and Computability and complexity.

Thank you for considering our manuscript. We look forward to the opportunity to contribute to Acta Informatica and we would be happy to provide any additional information or clarification as needed.

Sincerely,
Alexandre Silva